Inst Management & Operational Reserch

1. CPM & PERT (proj duration in CPM in very Imp)

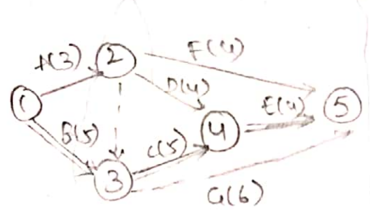
CPM

		Dummy	
Activity on Node (AON) :	O : Activity → connector,	✗	CPM
Activity on Arrow (AOA) :	→ : Activity O : connector (event)	✓	CPM & PERT

(All ways prefer this) (6 activites = 6 arrows)

Ex AOA

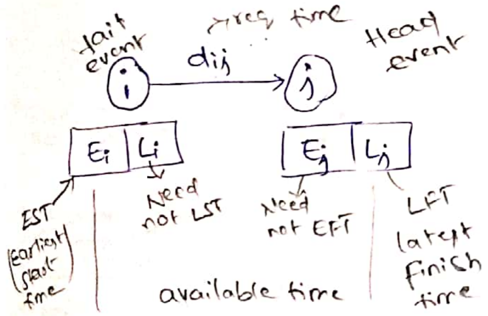
activity	precedor	dur
A	-	3
B	-	5
C	A,B	5
D	A	4
E	C,D	4
F	A	4
G	A,B	6



1-2-3-5 = 9
1-2-5 = 7
1-2-4-5 = 11
1-2-3-4-5 = 12
1-3-4-5 = 14
1-3-5 = 11

Ex AOA

Act	dur	precedence relationship	precedsor
A	5	A<B,C,D	–
B	7	B<E,F,G	A
C	10	C<I	A
D	6	D<G	A
E	3	E<H	B
F	9	F<I	B
G	7	G<I	B,D
H	4	H<I	E
I	2	---	C,F,G,H



Earliest Finish Time	(EFT) = EST + dur
Latest Start Time	(LST) = LFT - dur

Earliest occ time

Forward pass: (i) $\xrightarrow{d_{ij}}$ (j) $E_j = E_i + d_{ij}$

multiple path: (i) $\xrightarrow{d_{ik}}$ (k), (j) $\xrightarrow{d_{jk}}$ (k)

$E_k = \max\{E_i + d_{ik};\ E_j + d_{jk}\}$

Type of float	Equation	Conditions without affecting
Total Float (or) Float (TF)	$\boxed{TF = (L_j - E_i) - d_{ij}}$ (Available time − req time) $= LFT - EST - dur$ $= LFT - EFT$ $= LST - EST$	proj completion time
Free float (FF)	$\boxed{FF = TF - \text{Slack @ head event}}$ $= E_j - E_i - d_{ij}$	proj completion time Next activity
Independent float (IF)	$\boxed{IF = TF - \text{slack @ head} - \text{slack @ tail}}$ $= FF - \text{Slack}$ $= E_j - L_i - d_{ij}$	i. proj comp time ii. Next activity iii. preceding "

Latest occur time

Forward pass: (i) $\xrightarrow{d_{ij}}$ (j) $\boxed{L_i = L_j - d_{ij}}$

multiple path: (i) $\xrightarrow{d_{ij}}$ (j), (i) $\xrightarrow{d_{ik}}$ (k)

$L_i = \min\{L_j - d_{ij},\ L_k - d_{ik}\}$

Imp

Slacks: $E_j - L_j$ – Head event; $E_i - L_i$ – tail event

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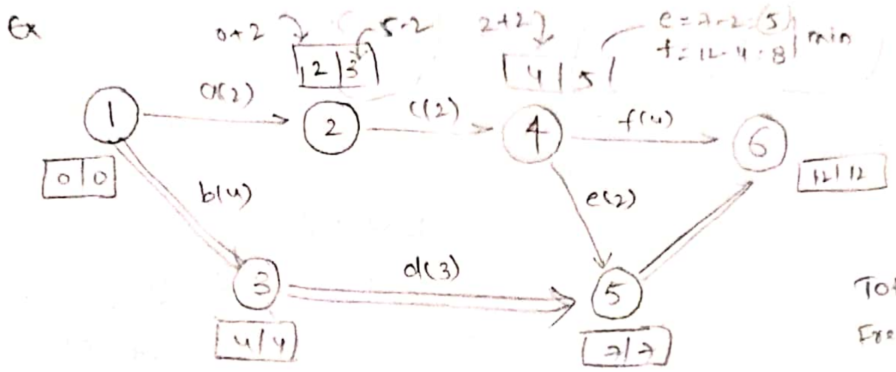
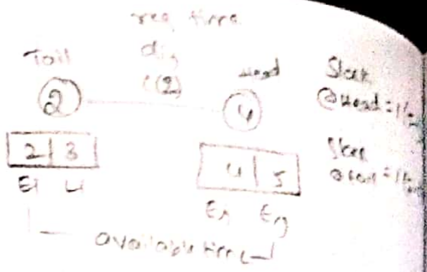
Ex





Total Float TF = (5-2) - 2 = 1

Free float = available time − req time = (TF) − slack @ head = 1 − 1 = 0

Independent float = TF − Slack head − slack tail = 1 − 1 − 1 = −1

PERT

Expected dur $t_E = \frac{t_o + 4t_m + t_p}{6}$ → optimistic time, most likely time, pessimistic time

V. Imp*

variance of single activity $V = \left(\frac{t_p - t_o}{6}\right)^2$

sd of critical path $\sigma_{cp} = \sqrt{V_{cp}}$ → variance of whole activities in the critical path

$V_{cp} = V_A + V_B \dots V_n$ → Critical path activities only (not all)

proj dur distribution $= \mu_{cp} \pm 3\sigma_{cp}$ → Critical path dur

$z = \frac{X - \mu_{cp}}{\sigma_{cp}}$

X → dur given in Question

PERT activity − β-distribution

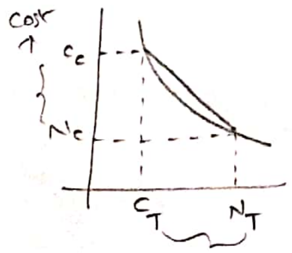
PERT proj dur − Normal distribution

V. Imp

Project crashing

V. Imp

$$\text{cost slope} = \frac{C_c - N_c}{N_T - C_T}$$

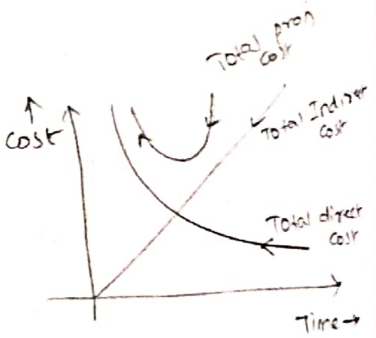


N_T = Normal time

N_c = Normal cost

C_T = crash time

C_c = Crash cost



Resource levelling

It is a process which the resources are allocated to proj activities without exceeding the specified max level.

critical path may getting delayed during the process

Resource smoothing

It is a process in which the resources are allocated in most uniform manner ie same amount of resources throughout proj dur

critical path remains same & No change is permitted

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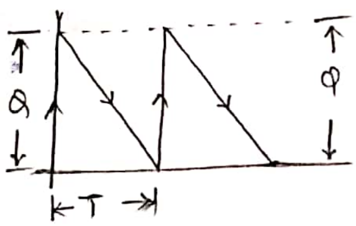
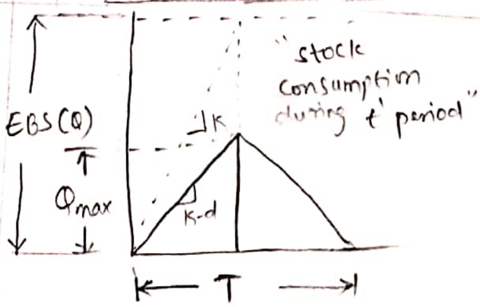
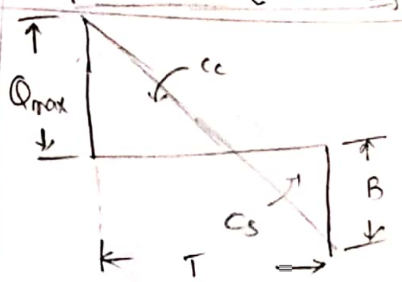
Inventory Control (2 or 3 m)

$\boxed{EOQ = Q}$ Imp

V.Imp Basic EOQ Model	Imp Mfg (or) production Model — EBS Model	Imp EOQ model with Shortage
V.Imp $\boxed{EOQ = \sqrt{\frac{2DC_0}{C_c}}}$	Economic Batch Size $EBS = \sqrt{\frac{2DC_0}{C_c}} \times \sqrt{\frac{K}{K-d}}$	$EOQ_s = \sqrt{\frac{2DC_0}{C_c}} \times \sqrt{\frac{C_c+C_s}{C_s}}$
Annual Inventory Cost (TIC) TIC = Annual Carring Cost + Annual ordering cost $TIC\|_Q = \frac{Q}{2} \times C_c + \frac{D}{Q} \times C_0$ $TIC\|_Q = \sqrt{2DC_0C_c}$	K = production rate/day d = demand/day C_0 = setup cost Belong to only EBS model $TIC\|_{EBS} = \frac{Q}{2}\left[\frac{K-d}{K}\right] \times C_c + \frac{D}{Q} \times C_0$ $TIC\|_{EBS} = \sqrt{2DC_0C_c}\sqrt{\frac{K-d}{K}}$	EOQ = Economic order Quantity D = Demand/year C_0 = ordering cost (Rs/order) C_c = carring cost (or) Holding cost (Rs/unit/year) $TIC\|_{EOQ_s} = \sqrt{2DC_0C_c}\sqrt{\frac{C_s}{C_c+C_s}}$ C_s = Back ordering cost (or) Shortage cost (or) Runout of stock cost
Total Annual cost (TC) $TC = TIC + (D \times C_u)$ C_u = unit cost of Item (Rs/unit)	$TC = \underbrace{TIC}_{\text{Annual Inventory cost}} + (D \times C_u)$	$TC = TIC + (D \times C_u)$
No. of orders $(N) = \frac{D}{EOQ}$	$N = \frac{D}{EBS}$	$N = \frac{D}{EOQ_s}$
Cycle Time $(T) = \frac{1}{N} = \frac{EOQ}{D}$	$T = \frac{1}{N}$ (year)	$T = \frac{1}{N}$
$Q_{max} = Q = EOQ$	$\boxed{Q_{max} = Q\left(\frac{K-d}{K}\right)}$	*V.Imp* $\boxed{Q_{max} = Q\left[\frac{C_s}{C_c+C_s}\right]}$







Note: In the above table Notations are same for any model

EOQ with price Breakdown (discount)

$$TC = \left[\frac{Q}{2} \times C_c + \frac{D}{Q} \times C_o\right] + (D \times C_u')$$

C_u' = unit Cost after discount

ex: 5% discount $Q = 100 \leq 200$

EOQ: $Q = 100$ $C_u = 50$/- unit

$C_u' = C_u - \text{discount}$ discount 5% of 50

$= 50 - 2.5$ $= 0.05 \times 50$

discount $= 2.5$

$\boxed{C_u' = 47.5}$

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Linear Programming

Possible solutions in Graphical method

1. unique optimal sol
2. multiple optimal sol or Alternative sol } Graph
3. Un-bounded Solution
4. Degenerate sol } observation we can get answer
5. Infeasible sol

Graphical method

$\leq$ - toward origin

$\geq$ away from origin

unique optimal: only have one sol and have the one corner point

multiple optimal: more than one corner point

maximization $\leq$ & minimization $\geq$ & parall constrain should satisfy $\Rightarrow$ multiple optimal sol

(But can't say fixely that if parallel satisfy then multiple opt. may not satisfy)

Ex: max $5x_1 + 7x_2 = 0$

$10x_1 + 14x_2 \leq 8 \rightarrow 5x_1 + 7x_2 \leq 4$

Both are parallel so may be multiple optimal

Un-Bounded Sol: maximization with Both variable upper limit ∞

un-Bounded Sol

Note: minimization case no need to use It is not at all un-Bounded Sol

Ex: max $z = 3x_1 + 2x_2$ → Imp variable b/c for max profit $3 > 2$

$-2x_1 + 3x_2 \leq 9$

$x_1 - 5x_2 \geq -20$

$-2x_1 \leq 9 \quad x_1 \geq -9/2$

$x_1 \geq -20$

Both these have the upper bound ∞ so there is un-Bounded sol

max $z = 3x_1 + 2x_2$

$2x_1 + x_2 \leq 8 \Rightarrow 2x_1 \leq 8 \Rightarrow x_1 \leq 4$ → have upper Bound (not ∞)

$5x_1 + 2x_2 \geq 18 \Rightarrow 5x_1 \geq 18 \Rightarrow x_1 \geq \frac{18}{5}$ upper bound ∞

If one condition failed then it is an not unbounded Solution

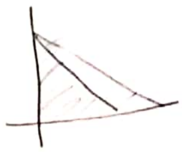
Degenerate possible for maximization & minimization, if we got same values for constraints for imp variable then such cases degenrate occurs

degenerate is a special case in unique optimal sol

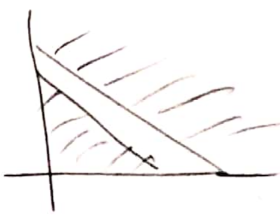
Ex: $z = 5x_1 + 3x_2$ → maximization Max profit $5 > 3$ so it is imp variable

Imp variable $2x_1 + 3x_2 \leq 4 \qquad 2x_1 \leq 4 \Rightarrow x_1 \leq 2$

$3x_1 + 4x_2 \leq 6 \qquad 3x_1 \leq 6 \Rightarrow x_1 \leq 2$ — same so it is degenrate



Infeasible sol: no common (feasible) region



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	products		
	product I	product II	RHS
Resources R1	per unit requirement		
R2			
R3			
cost/unit			

Simplex Method

n-variables & $\leq$ type constraints with non-negative RHS

ex: $\max z = 2000x_1 + 300x_2$

$3x_1 + 2x_2 \leq 90$

$x_1 + 2x_2 \leq 100$

$x_1 \geq 0,\ x_2 \geq 0$

Std LP: $\max z = 2000x_1 + 3000x_2 + 0S_1 + 0S_2$

$z - 2000x_1 - 3000x_2 - 0S_1 - 0S_2 = 0$

$3x_1 + 2x_2 + S_1 + 0S_2 = 90$

$x_1 + 2x_2 + 0S_1 + S_2 = 100$

In maximisation least -ve integer → entering variable

Row	Basic	x_1	x_2	S_1	S_2	RHS	Ratio
R_0	z	-2000	-3000	0	0	0	
R_1	S_1	3	2	1	0	90	$\frac{90}{2} = 45$
R_2	S_2	1	2	0	1	100	$\frac{100}{2} = 50$

key row; least +ve ratio

need to change → 0

leaving variable

key column

variable

Row	Basic	x_1	x_2	S_1	S_2	RHS	Ratio
$R_0 + 3000R_1'$	z	$-2000 + 3000(\frac{3}{2}) = 2500$	$-3000 + 3000(1) = 0$	$0 + 3000(\frac{1}{2}) = 1500$	$0 + 3000(0) = 0$	$0 + 3000(45) = 135000$	
$\frac{R_1}{2} = R_1'$	x_2	$3/2$	$2/2 = 1$	$\frac{1}{2}$	$\frac{0}{2} = 0$	$\frac{90}{2} = 45$	
$R_2 - 2R_1'$	S_2	$1 - 2(\frac{3}{2}) = -2$	$2 - 2(1) = 0$	$0 - 2(\frac{1}{2}) = -1$	$1 - 2(0) = 1$	$100 - 2(45) = 10$	

All z-row elements are non-negative = optimal solution

No. of zero elements in z row = 2

No. of basic variable = 2

} unique optimal

		x_1	x_2	S_1	S_2	RHS
R_0'	z	2500	0	1500	0	1,35,000
R_1'	x_2	$3/2$	1	$1/2$	0	45
R_2'	S_2	-2	0	-1	1	10

No -ve integer in the z row so it is optimal solution we can stop doing table

It is not consider as 0 in row-z

$z_{max} = 1,35,000$

$x_2 = 45$

$S_2 = 10$

Basic variables x_2, S_2

Non Basic variables $x_1 = 0$, $S_1 = 0$

Unique optimal sol: No. of zero elements in z-row = No. of basic variables

Multiple optimal sol: No. of zero elements in z-row > No. of basic variables

Imp variable

$\max z = 5x_1 + 3x_2 \left\{ \begin{array}{l} 2x_1 + 3x_2 \le 6 \Rightarrow \frac{6}{2} = 3 \\ 3x_1 + 4x_2 \le 9 \Rightarrow \frac{9}{3} = 3 \end{array} \right.$

Degenerate sol: If least +ve ratio's equal tie (If one Basic variable zero the ratio degenerate

Unbounded sol: If there is No least +ve ratio

$\max z = 3x_1 + 2x_2$

No +ve ratio

$\left. \begin{array}{l} -2x_1 + 3x_2 \le 9 \Rightarrow 9/-2 \\ -x_1 + 5x_2 \le 20 \Rightarrow 20/-1 \end{array} \right\}$

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Simple queuing Model

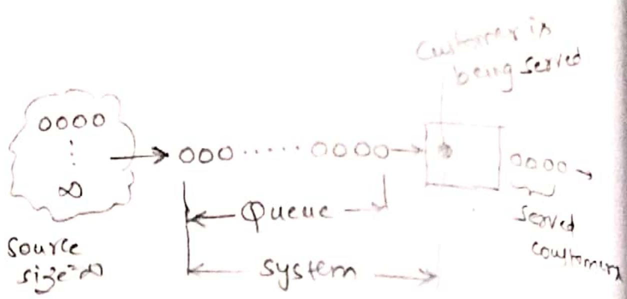
Arrival pattern $\boxed{\lambda = \frac{1}{\text{mean arrival time}}}$ mean arrival rate

$\lambda = \frac{\text{No. of arrival}}{\text{time}}$

Eg: arrival time 10 min

$\lambda = \frac{1}{\text{arrival time}} = \frac{1}{10\,min}$

$= \frac{1}{10 \times \frac{1}{60}} = \frac{60}{10}$ /hr



Serive pattern

$\mu = \frac{1}{\text{mean service time}}$ /hr

mean serive rate

μ & λ should be same units

V Imp

max units all should be in hrs

V Imp

No. of servers $\boxed{\delta = \frac{\lambda}{\mu}}$

$\delta < 1$ single server

$\delta \geq 1$ multi server

Single server Queuing model

- No. of arrivals per unit time will follows "poission distribution"
- Inter arrival time (or) mean arrival time follows "Exoponential distribution"

- Working time of server $= \rho \times$ available time → given in Q/S like 8 hr shift
- Idle time of server $= (1-\rho) \times$ available time
- Idle time cost per hr $= L_s \times$ Idle time cost /unit/hr (Rs/unit/hr) + Boy charge Repair
- Idle time cost (in Rs) $= W_s \times$ Idle time cost per hr

V. Imp

$$\rho = \frac{\lambda}{\mu}$$

$$L_s = \frac{\rho}{1-\rho}, \quad L_q = \rho \cdot L_s$$

$$W_s = \frac{L_s}{\lambda}, \quad W_q = \rho \cdot W_s$$

$$P(n) = \rho^n (1-\rho)$$

ρ = utilization factor

L_s = length of system

L_q = length of queue

W_s = waiting time of system

W_q = waiting time in queue

Poission distribution

$$P(n) = \frac{\lambda^n e^{-\lambda}}{n!}$$

$$\boxed{P(n,t) = \frac{(\lambda t)^n e^{-(\lambda t)}}{n!}}$$

Ex: No coustomers for first 3min prop that coustomers for next 3min

$P(n=0, t=3)$ = pro no customer in first 3 min {prop next 3min $= 1 - P(n=0, t=3)$

(Inter arrival time) ⇒ Exponential distribution

$$f(t) = \lambda e^{-\lambda t}$$

$$F(t) = \int_0^t f(t)\,dt$$

$$\boxed{F(t) = (1 - e^{-\lambda t})}$$

Ex: prop that time b/w successive coustomers (no 3 min)

$F(3) - F(1)$

$F(t) = 0$ to t $\quad f(t) = $ at t

$P(n=0) = 1-\rho$ $\quad$ n = No. of coustomers.

$P(n \neq 0) = \rho$

$P(n \geq k) = \rho^k$ (at least k coustomers)

$P(n > k) = \rho^{k+1}$

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Sequencing and Scheduling

most imp priority rules

1. shortest processing time (SPT)
2. Earliest Due date (EDD)

n - Jobs - 1 machine

Job	P	Q	R	S
processing time (days)	12	9	21	10
due date (days)	20	40	30	19

	$\bar{F}$	$\bar{T}$
SPT	27.75	8.25
EDD	31.75	6.75

EDD

S-P-R-Q

processing time (t_j), flow time (F_j), due date (d_j), Lateness ($L_j = F_j - d_j$), Tardiness ($T_j = \max\{0, L_j\}$)

Job	t_j	F_j	d_j	$L_j = F_j - d_j$	$T_j = \max\{0, L_j\}$
S	10	10	19	10-19 = -9	0
P	12	10+12=22	20	22-20 = 02	2
R	21	22+21 = 43	30	13	13
Q	09	43+9 = 52	40	12	12

$N_T = 3$ No of delays

$\Sigma F_j = 127$ $\qquad \Sigma T_j = 27$

mean flow time $\bar{F} = \frac{127}{4} = 31.75$

Mean Tardiness $\bar{T} = \frac{27}{4} = 6.75$

SPT rule (If any two same means, take min due one)

Job seq Q-S-P-R

Job	t_j	F_j	d_j	$L_j = F_j - d_j$	$T_j = \max\{0, L_j\}$
Q	9	9	40	-31	0
S	10	19	19	0	0
P	12	31	20	11	11
R	21	52	30	22	22

$\Sigma F_j = 111$ $\qquad \Sigma T_j = 33$

$\bar{F} = \frac{111}{4} = 27.75$ $\qquad \bar{T} = \frac{33}{4} = 8.25$

n - Jobs - 2 machines → imp

Car	Denting	painting
1	4	3
2	10	2
3	2	5
4	05	04
5	6	02
6	01	06

In this case any thing we can take our wish

car	denting	painting
2	10	2
3	2	5
5	6	2

Job (car)	Denting In	Denting PT	Denting out	painting In	painting PT	painting out
6	0 +	1 →	1	1 +	2 →	3
3	1 +	2 →	3	3 +	5	8
4	3 +	5 →	8	8 +	4	12
1	8	4	12	12	3	15
2	12	10	22	22	2	24
5	22	6	28	28	2	$\boxed{30}$ make span

which is big we should write that

denting | 6 | 3 | 4 | 1 | 2 | 5 |

Actual working time → 28

available time of denting

22

imp

Idle time of denting = 28 - 28 = 0

Idle time of painting = 30 - 22 = 8

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Forecasting

1. Simple moving Average method – 1m q/s

Ex:

month	Demand	
march	460	three-month moving avg method given in q/s
April	510	
may	520	
June	495	3 month moving avg
July	475	
Aug	560	
sep	?	

** V.Imp **

forecast for sep

$$\boxed{F_{sep} = \frac{D_{Aug}+D_{July}+D_{June}}{3}}$$

$= \frac{560+475+495}{3} = 510$ units

2. Exponential Smoothing – 2m q/s

** V.Imp **

$$\boxed{F_{t+1} = F_t + \alpha(D_t - F_t)}$$

α → Smoothing coefficient

Ex:

month	Demand	Forecast
June	300	200
July	350	
Aug		?

$\alpha = 0.7$ → given in q/s

$F_{July} = 200 + 0.7(300-200) = 270$

forecast for Aug $F_{Aug} = F_{July} + \alpha(D_{July} - F_{July})$

$= 270 + 0.7(350-270) = 326$ Ans ← F_{Aug}

$$\boxed{\alpha = \frac{2}{n+1}}$$ $0 \le \alpha \le 1$

Stable demand $n\uparrow \alpha\downarrow$

Unstable $n\downarrow \alpha\uparrow$

α : for recent demand

Forecast errors

Mean forecast error $\boxed{(MFE) = \frac{\Sigma(D_t - F_t)}{n}}$ n → no. of months

Mean absolute error $\boxed{(MAD) = \frac{\Sigma|D_t - F_t|}{n}}$

Mean Square Error $\boxed{(MSE) = \frac{\Sigma(D_t - F_t)^2}{n}}$

Mean absolute percentange Error (MAPE) $= \boxed{\frac{1}{n}\left[\frac{\Sigma|D_t - F_t|}{D_t} \times 100\right]}$

Sum of absolute deviations $= \boxed{\Sigma|D_t - F_t|}$